

Redesigning DATA-R155 for the AI Era

Specifications Grading, AI Competencies, & Reflections



Lorraine Gaudio

Adjunct Instructor | Anthropology
RISE Fellow | PhD in Computing

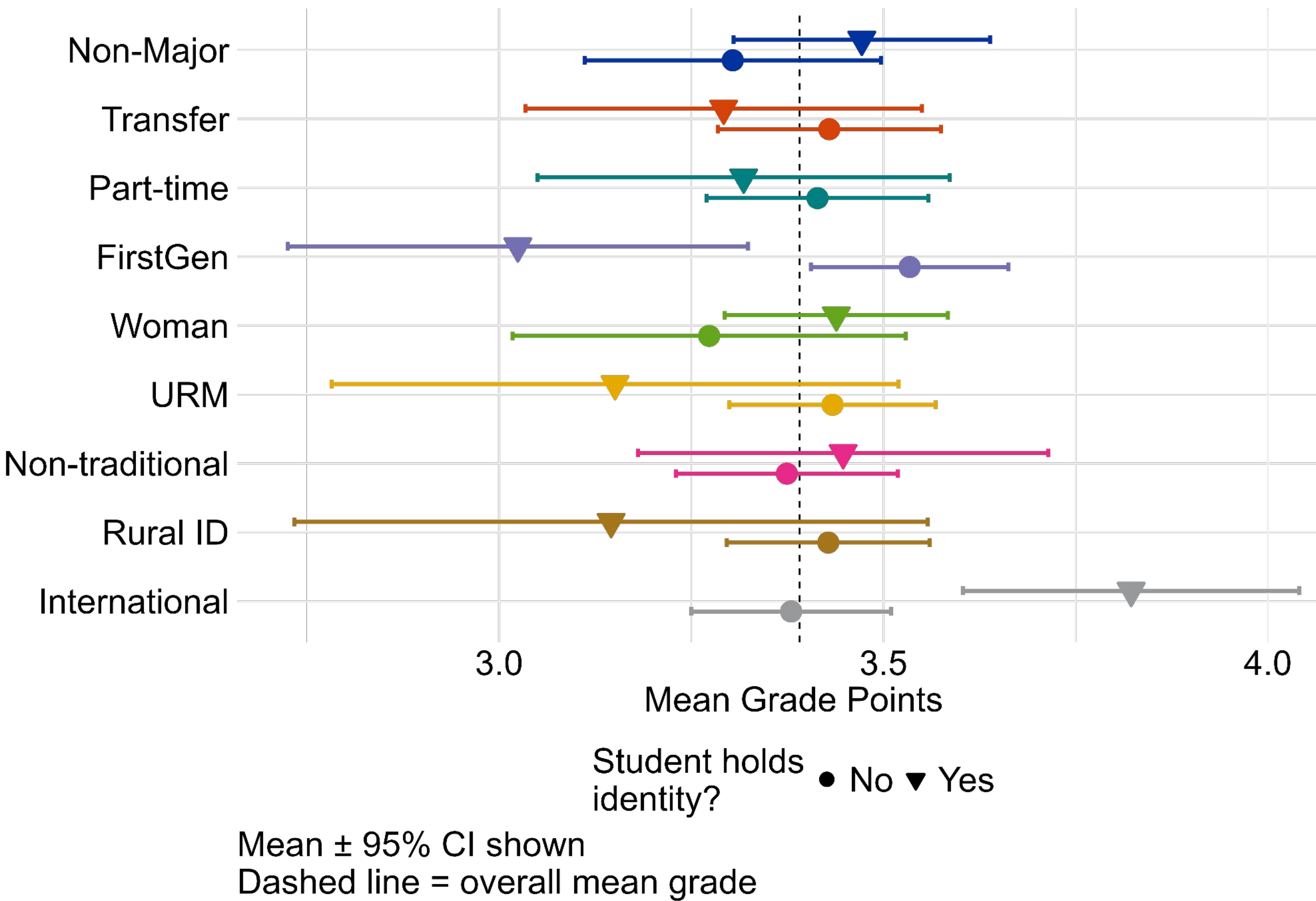
Introduction

LLMs may transform how new knowledge is produced, making it necessary for future researchers to acquire more than technical skills. Yet LLMs change what students can produce, creating a risk that performance can outpace learning. Courses need new designs that make learning visible¹, evaluable², and responsibly guided³. In a redesigned 7-week introductory R programming course, I use specifications grading, self-regulated learning memos, and AI competency training. This study examines whether the redesign improves course success, R Certificate continuation, and effective learning routines compared with prior offerings and a comparable course, with attention to equity-related gaps.

Baseline Equity Gaps

Historical baseline patterns show uneven mean grade points. The gap between first and continuing generation showed the largest gap. Underrepresented minorities and rural Idaho students were also below the overall mean. Similar, yet less extreme, patterns appeared in the control course (not disclosed). “Major” = identified for R Certificate.

These pre-treatment gaps motivated redesign decisions to improve clarity, support, and persistence.



Study Design & Measures

Comparative Mixed-Methods

1. Assess if and how the redesign changes student outcomes
2. Measure changes in R workflow, learning practice, and AI competencies
3. Examine course reflection artifacts for evidence of learning

Study Design

- **Treatment Group:** DATA-R155 students Spring 2026-Spring 2028
- **Historical Baseline:** DATA-R155 students Fall 2020-Fall 2025
- **Control Course:** benchmark broader trends not specific to the redesign

Key Measures

- **Course Outcomes:** grades D, F, or withdrawal; disaggregated data from treatment, baseline, and comparison course
- **Program Continuation:** declared R Certificate students remaining on the R certificate path comparing treatment with baseline
- **Pre/Post Treatment Course Survey:** R workflow familiarity, learning habits, and AI competencies; Likert scale items
- **Treatment Course Artifacts:** student and instructor reflections

Course Redesign Features

Specifications grading

core and advanced activities in grade bundle (A, B, C) pathways

Learning memos

self-regulated learning model: forethought → performance → self-reflection

AI competencies

calculated, human-centered, transparent use decisions for learning and research

Workflow supports

redesigned Canvas, clear weekly schedule, guided practice, and support plan

7-Week

shortened from full term, with in person and online options

Iterative design

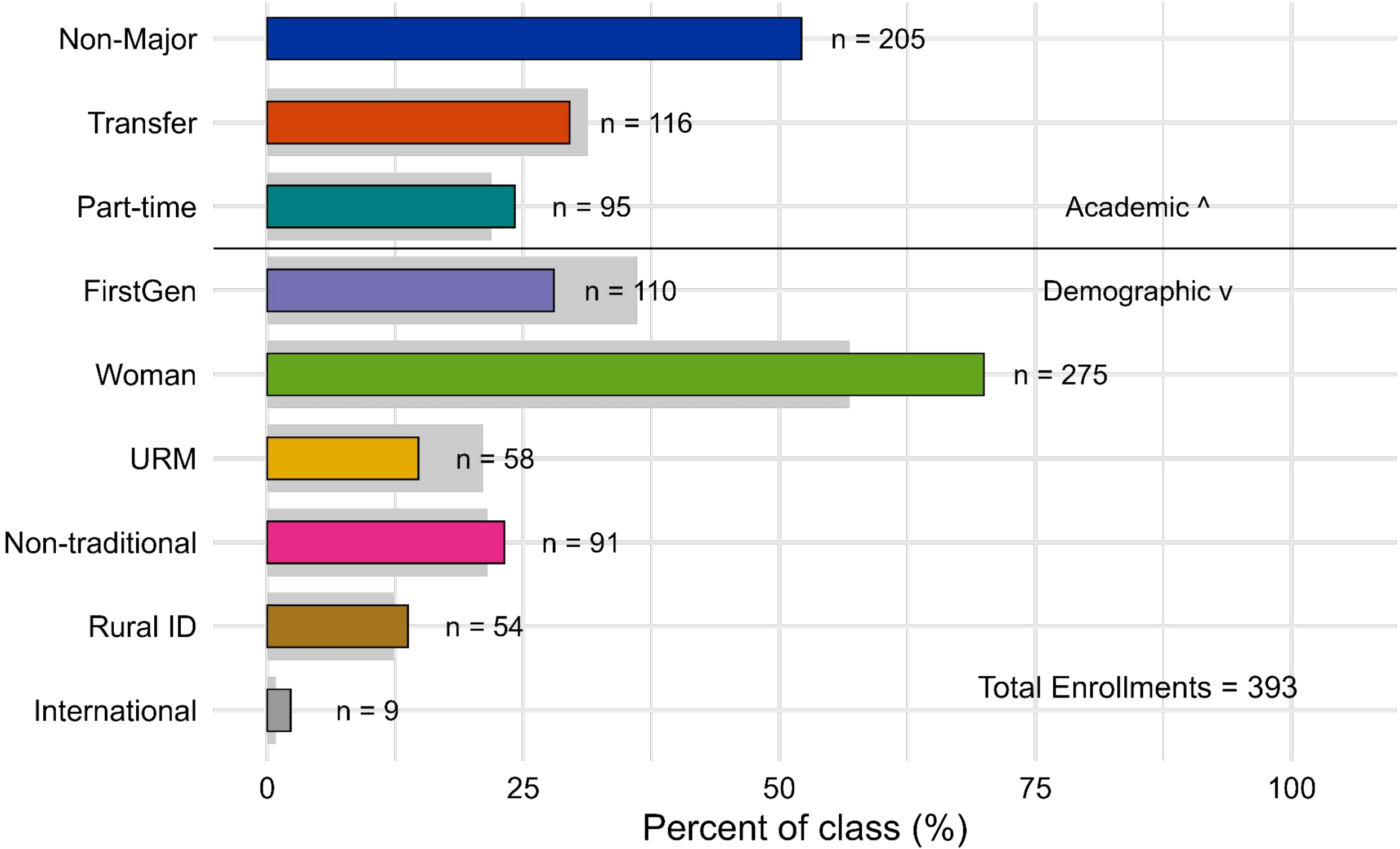
multi-stakeholder co-design process



Who the Course Serves

- **STEM and Social Science Majors:** Cross-listed ANTH155, BIOL155, PSYC155, and SOC155.
- **Mixed academic population:** freshman (9%), sophomore (18%), juniors (32%) and seniors (23%), graduate students (14%) and post-bacc students (4%).

Baseline demographic patterns show how DATA-R155 enrollment compares with the broader university population. Color = DATA-R155; gray = Boise State. Identities are not mutually exclusive.



Potential Implications

If this study supports the redesign, it will inform the design of AI-era coding courses on designs to make learning visible, equitable, and student-centered.

- Clear mastery standards may improve student success
- Equity-focused course design helps reduce persistent gaps
- Teaching AI competencies may strengthen learning
- Iterative co-design supports responsive teaching
- Offers a scalable model

References

1. Paludo & Montresor (2024). Fostering Metacognitive Skills in Programming: Leveraging AI to Reflect on Code.
2. Nilson & Packowski (2025). *Specifications Grading 2.0: Restoring Rigor, Motivating Students, Saving Faculty Time, and Developing Career Competencies*.
3. Kisselburgh & Beever (2022). The Ethics of Privacy in Research and Design. In *Modern Socio-Technical Perspectives on Privacy*